MASTER_PLAN.md

SynChain

AI-Powered Supply Chain Decision Intelligence Platform

Version: 1.0

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1. Product Vision

What is SynChain?

SynChain is a Multi-Agent AI Supply Chain Digital Twin platform that helps organizations make better operational decisions before executing them in the real world.

Instead of asking:

"How much inventory do I have?"

SynChain answers:

"What should I do next?"

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2. Problem Statement

Current supply chain systems are reactive.

Most organizations:

- Analyze historical data
- Use spreadsheets
- Depend on manual planning

Problems:

- Demand uncertainty
- Excess inventory
- Stock shortages
- Supplier delays

- Logistics inefficiencies

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3. Solution

SynChain creates a virtual representation of a supply chain.

This virtual representation is called:

Digital Twin

Specialized AI agents analyze the Digital Twin and generate recommendations.

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4. Product Goals

Primary Goals:

- Forecast demand
- Optimize inventory
- Reduce operational risk
- Improve logistics planning
- Simulate future scenarios

Secondary Goals:

- Explain recommendations
- Monitor external events
- Improve decision quality

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5. Complete System Architecture

User ↓ Frontend ↓ API Layer ↓ Orchestrator Agent ↓ News Agent Market Agent Supplier Agent ↓ Demand Agent ↓ Inventory Agent ↓ Logistics Agent ↓ Risk Agent ↓ Scenario Agent ↓ Recommendation Agent ↓ Explanation Agent ↓ Dashboard

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6. Technology Stack

Frontend:

- Next.js
- React
- TypeScript
- TailwindCSS
- Shadcn UI

Backend:

- FastAPI
- Python

Database:

- PostgreSQL (Supabase)

AI:

- LangGraph
- Ollama
- Qwen / DeepSeek

Search:

- Tavily
- GDELT
- RSS

Deployment:

- Vercel
- Render

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7. Development Phases

================================= PHASE 1 FOUNDATION =================================

Goal:

Build working platform.

Tasks:

Frontend

- Landing Page
- Form Page
- Results Page

Backend

- FastAPI Setup
- PostgreSQL Connection

Database

- Schema Creation

Deliverable:

Working simulation flow.

================================= PHASE 2 AGENT SYSTEM =================================

Goal:

Implement AI agents.

Tasks:

Demand Agent

Inventory Agent

Logistics Agent

Risk Agent

Recommendation Agent

Explanation Agent

Deliverable:

Agent pipeline operational.

================================= PHASE 3 DIGITAL TWIN =================================

Goal:

Create virtual supply chain.

States:

Inventory State

Warehouse State

Supplier State

Demand State

Transportation State

Market State

Risk State

Deliverable:

Agents operate on Digital Twin.

================================ PHASE 4 SCENARIO ENGINE
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Goal:

What-if simulation.

Scenarios:

Demand Surge

Demand Decline

Supplier Failure

Warehouse Closure

Transport Delay

Deliverable:

Scenario Simulator.

================================= PHASE 5 EXTERNAL INTELLIGENCE
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Goal:

Integrate real-world signals.

Sources:

Tavily

GDELT

RSS

Deliverable:

External signals influence decisions.

================================= PHASE 6 ENTERPRISE FEATURES
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Goal:

Multi-user platform.

Features:

Authentication

Workspaces

Reports

Analytics

Deliverable:

Enterprise-ready system.

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8. Agent Implementation Plan

Agent 1

Demand Agent

Input:

- Product
- Demand Data
- Market Signals

Output:

Forecast

Confidence

Reasoning

Agent 2

Inventory Agent

Input:

- Current Stock
- Demand Forecast

Output:

Inventory Recommendation

Safety Stock

Agent 3

Logistics Agent

Input:

- Warehouse Data

Output:

Warehouse Recommendation

Agent 4

Risk Agent

Input:

- Supplier Data
- Logistics Data

Output:

Risk Score

Agent 5

News Intelligence Agent

Input:

- News Sources

Output:

Event Signals

Impact Score

Agent 6

Market Intelligence Agent

Input:

- Industry Trends

Output:

Market Signals

Agent 7

Supplier Intelligence Agent

Input:

- Supplier Information

Output:

Reliability Score

Agent 8

Scenario Agent

Input:

- Digital Twin

Output:

Future Outcomes

Agent 9

Recommendation Agent

Input:

All Agent Outputs

Output:

Final Strategy

Agent 10

Explanation Agent

Input:

Recommendations

Output:

Human-readable Explanation

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9. Database Implementation Order

Step 1

users

Step 2

companies

Step 3

products

Step 4

warehouses

Step 5

suppliers

Step 6

simulations

Step 7

digital_twin_states

Step 8

agent_outputs

Step 9

recommendations

Step 10

news_signals

Step 11

risk_events

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10. API Implementation Order

1. GET /health
2. POST /simulate
3. GET /simulation/{id}
4. POST /simulation/{id}/run
5. GET /simulation/{id}/results
6. GET /simulation/{id}/agents
7. POST /simulation/{id}/scenario
8. GET /news
9. GET /recommendations

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11. Frontend Implementation Order

Page 1

Landing Page

Sections:

Hero

How It Works

Features

CTA

Footer

Page 2

Simulation Form

Inputs:

Product

Stock

Demand

Supplier Delay

Warehouse

Page 3

Results Dashboard

Cards:

Demand Forecast

Inventory Recommendation

Logistics Plan

Risk Analysis

Final Recommendation

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12. Success Criteria

MVP Success:

User enters data.

Agents execute.

Digital Twin updates.

Recommendations generated.

Results displayed.

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13. Future Expansion

Supplier Scoring

Predictive Procurement

ERP Integration

IoT Integration

Carbon Optimization

Autonomous Supply Chain

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14. Long-Term Vision

SynChain becomes:

"AI Operating System for Supply Chain Decision Intelligence"

Organizations no longer use SynChain only for simulation.

They use it to make daily operational decisions.